Requirements Analysis & System Design

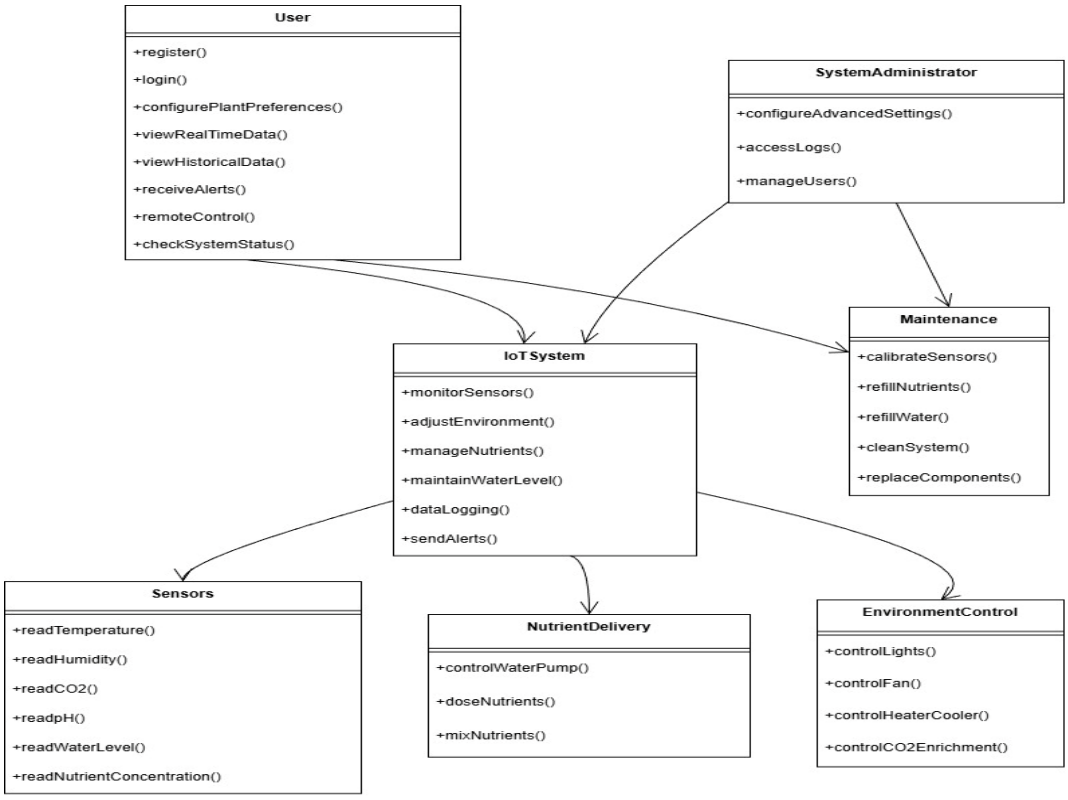
Complete Use Case Diagrams and Analysis

System Overview

IoT-powered hydroponic system for growing leafy greens in South Africa, featuring automated nutrient delivery, environmental control, and remote monitoring via ESP32.

Actor Identification

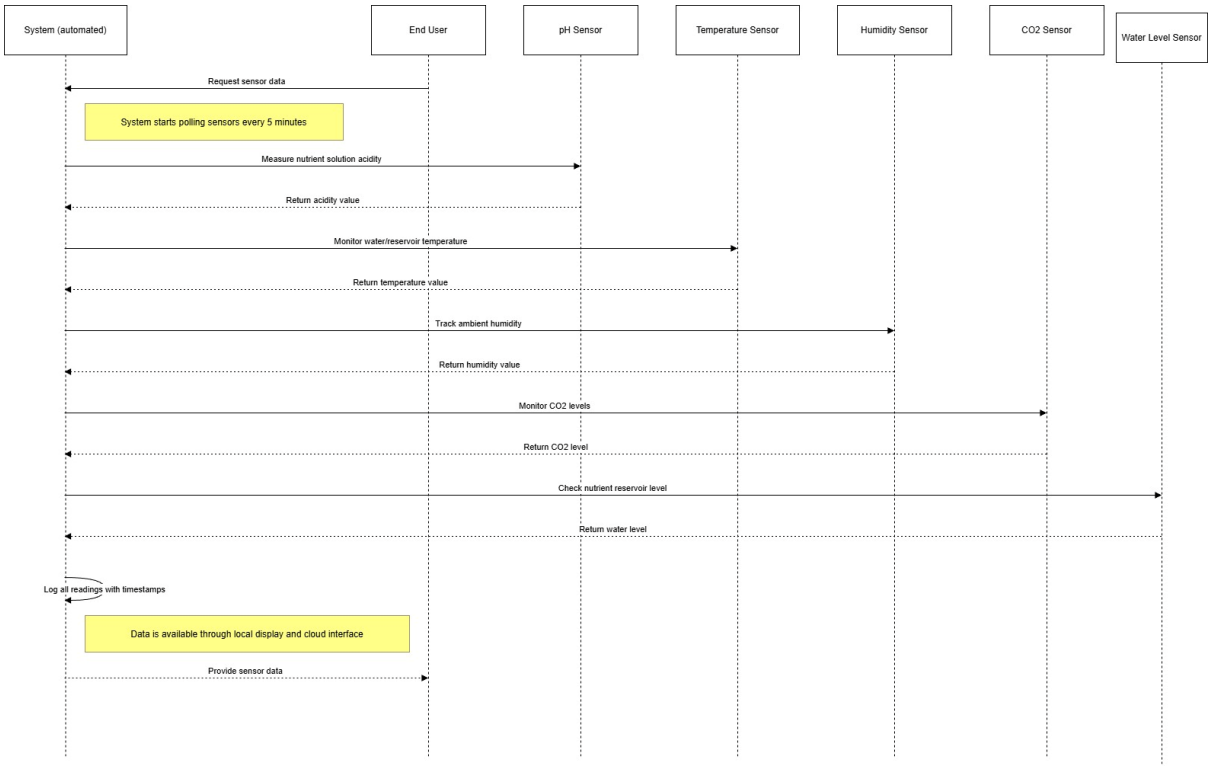
Primary Actors	Secondary Actors
End User - Home grower using the system	**Cloud Service** - External IoT platform for data storage/processing
Mobile App User - User accessing system remotely	**Alert System** - Sends notifications
System Administrator - Advanced user/developer (you)	**Plant Database** - Stores plant-specific requirements
	Calendar/Time Service - Provides scheduling functions



UC-1: Monitor Environmental Conditions

Primary Actors	Secondary Actors	Preconditions	Main Flow	Alternate Flow	Post Conditions
System (automated)	End User	System powered on, sensors calibrated	System continuously polls all sensors at 5-minute intervals	System detects sensor reading outside expected range	Environmental data stored and available for analysis
			pH sensor measures nutrient solution acidity (target: 5.5-6.5 for leafy greens)	Marks sensor as "needs calibration"	
			Temperature sensor monitors water/reservoir temperature (target: 18-22°C)	Sends alert to user	
			Humidity sensor tracks ambient humidity (target: 40-60% for lettuce/spinach)	Continues operation with remaining sensors	

			CO2 sensor monitors CO2 levels (target: 400-800 ppm)		
			Water level sensor checks nutrient reservoir level		

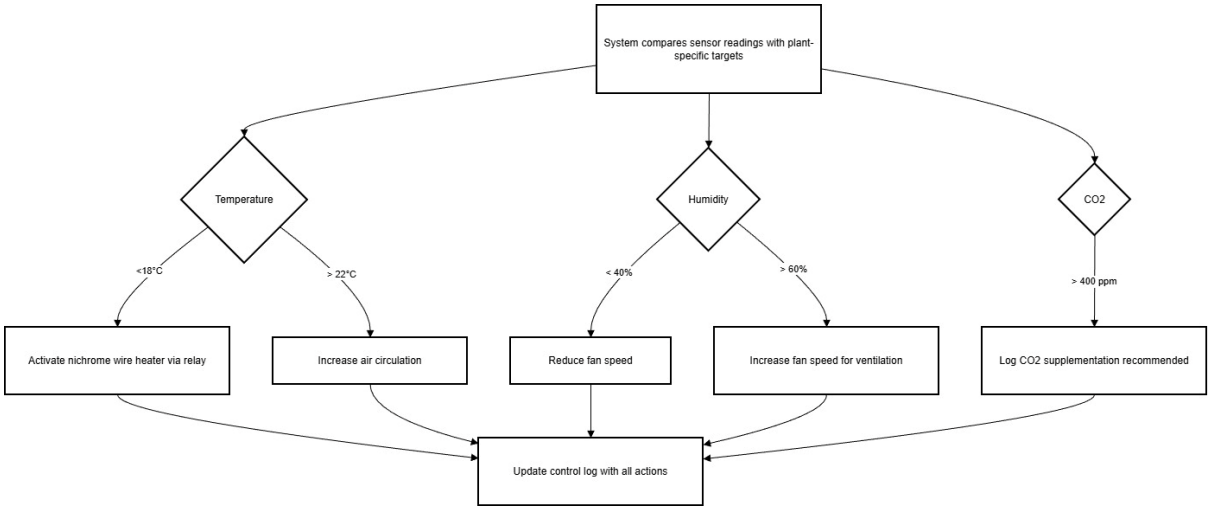


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UC-2: Control Environmental Conditions

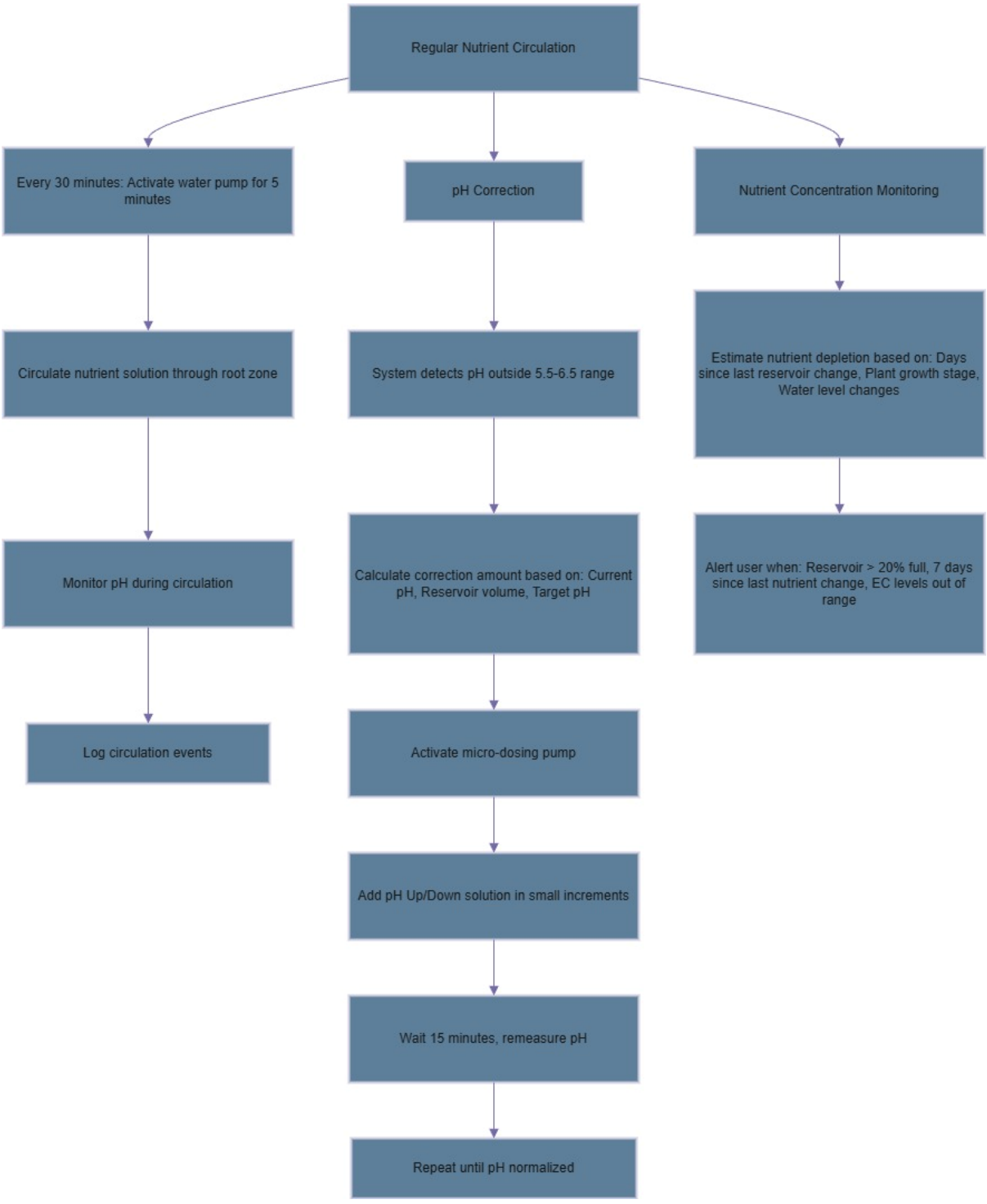
Primary Actors	Secondary Actors	Preconditions	Main Flow	Alternate Flow	Post Conditions
System (automated)	End User	Environmental parameter deviates from setpoint	System compares sensor readings with plant-specific targets	User remotely adjusts temperature setpoint	Environment maintained within target ranges
			If temperature < 18°C), Activate nichrome wire heater via relay	System acknowledges override	
			If temperature > 22°C: Activate axial fan for cooling	Operates according to manual settings	
			If temperature > 22°C: Activate axial fan for cooling		

			If humidity < 40%: Reduce fan speed		
			If CO2 < 400 ppm: Note for future CO2 enrichment capability		



UC-3: Manage Nutrient Delivery System

Row	Actor	Trigger / Condition	Steps / Logic
1	System	Scheduled cycle (every 30 minutes)	1. Activate water pump for 5 minutes. 2. Circulate nutrient solution through root zone. 3. Monitor pH during circulation. 4. Log circulation events.
2	System	Detects pH outside optimal range (5.5 - 6.5)	1. Calculate correction amount based on current pH, reservoir volume, and target pH. 2. Activate micro-dosing pump (future enhancement).
3	System	During pH Correction	3. Add pH Up/Down solution in small increments. 4. Wait 15 minutes, then remeasure pH. 5. Repeat steps until pH is normalized.
4	System	Scheduled monitoring & estimation	1. Estimate nutrient depletion based on: days since last change, plant growth stage, and water level changes.
5	System	Meets alert condition	2. Alert user when: reservoir is <20% full, 7 days since last nutrient change, or EC levels are out of range (if sensor available).
6	End User	Manual intervention / adjustment	Alternate Flow: 1. User adds nutrients manually. 2. User logs amount added via app. 3. System recalculates depletion timeline. Postcondition: Plants receive adequate nutrients at proper pH.


Regular Nutrient Circulation
Every 30 minutes: Activate water pump for 5 minutes
Circulate nutrient solution through root zone
Monitor pH during circulation
Log circulation events
pH Correction
System detects pH outside 5.5-6.5 range
Calculate correction amount based on: Current pH, Reservoir volume, Target pH
Activate micro-dosing pump
Add pH Up/Down solution in small increments
Wait 15 minutes, remeasure pH
Repeat until pH normalized
Nutrient Concentration Monitoring
Estimate nutrient depletion based on: Days since last reservoir change, Plant growth stage, Water level changes
Alert user when: Reservoir > 20% full, 7 days since last nutrient change, EC levels out of range

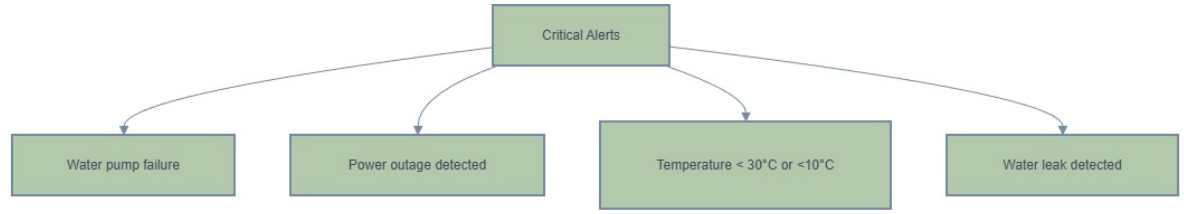
UC-4: Alert User of Issues

Alert Conditions & Categories

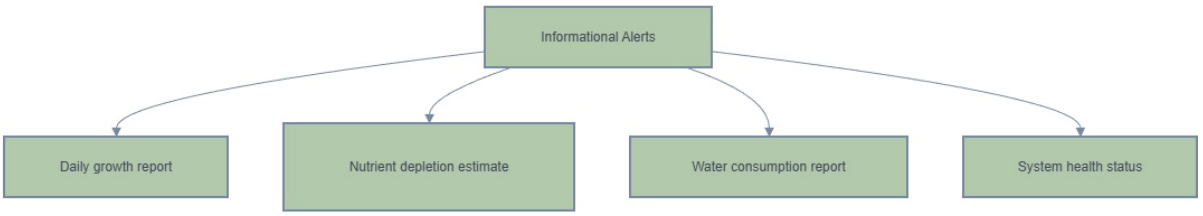
Alert Level	Condition / Trigger	Examples
Primary Actor	System	
Secondary Actor	End User, Alert System	
Critical Alerts	Immediate notification required.	• Water pump failure • Power outage detected • Temperature >30°C or <10°C • Water leak detected (via moisture sensor)
Warning Alerts	Hourly notification if unresolved.	• pH out of range for >2 hours • Nutrient reservoir <20% full • Any sensor malfunction • WiFi connection lost
Informational Alerts	Daily summary notification.	• Daily growth report • Nutrient depletion estimate • Water consumption report • System health status

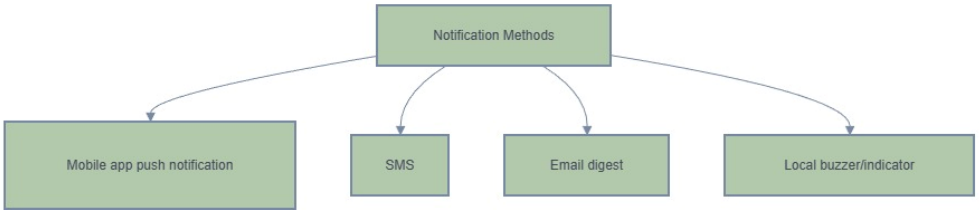
Notification Methods

Notification Method	Description / Use Case
Mobile App Push Notification	Primary method for all alert levels.
SMS	Used for critical alerts, especially relevant for users in South Africa.
Email Digest	Used for informational alerts and summaries.
Local Buzzer / Indicator	Provides immediate, on-site attention for critical hardware issues.


Critical Alerts
Water pump failure
Power outage detected
Temperature < 30°C or <10°C
Water leak detected

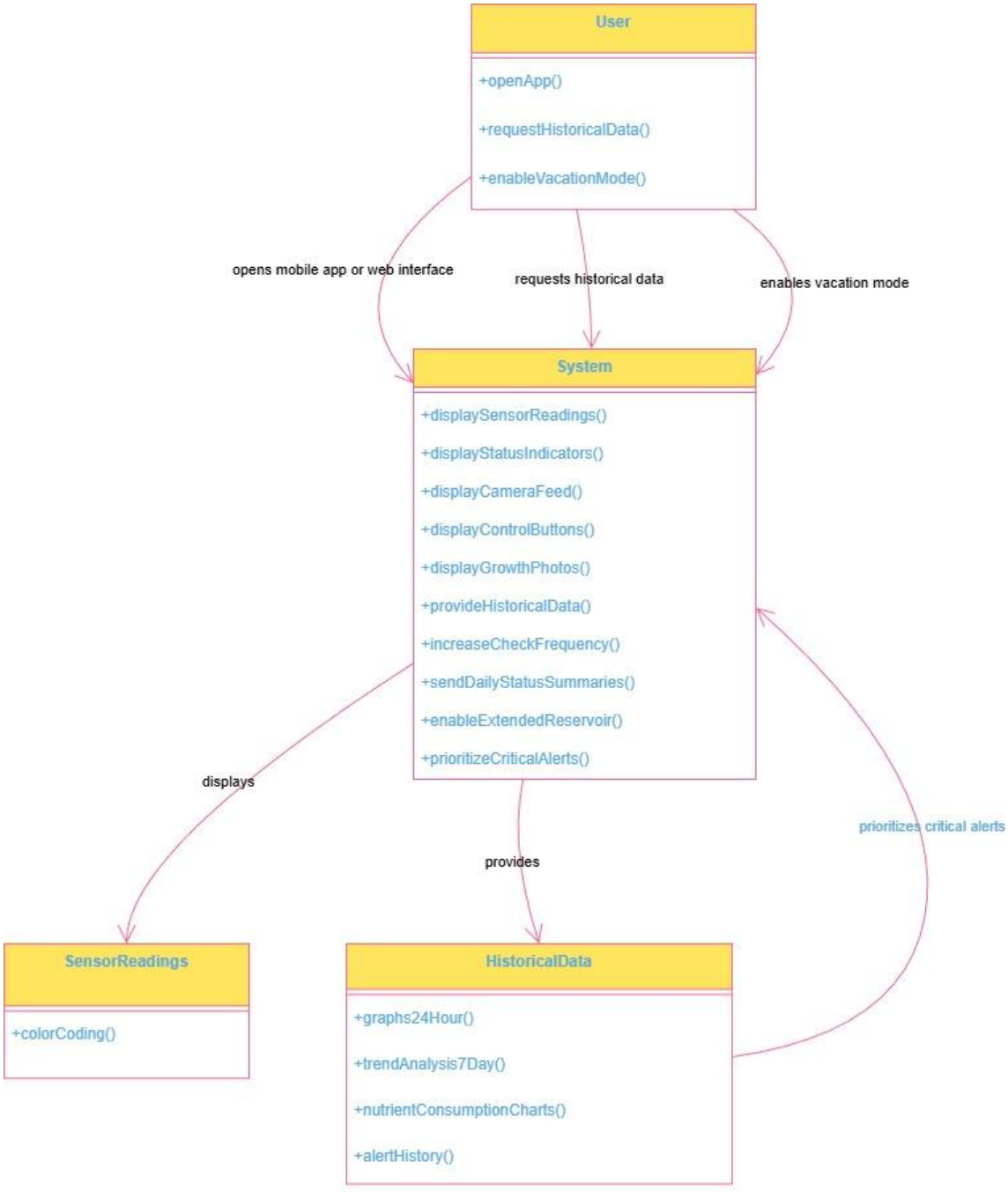

Warning Alerts
pH out of range for > 2 hours
Nutrient reservoir > 20%
Any sensor malfunction
WiFi connection lost


Informational Alerts
Daily growth report
Nutrient depletion estimate
Water consumption report
System health status


Notification Methods
Mobile app push notification
SMS
Email digest
Local buzzer/indicator

UC-5: Report System Status

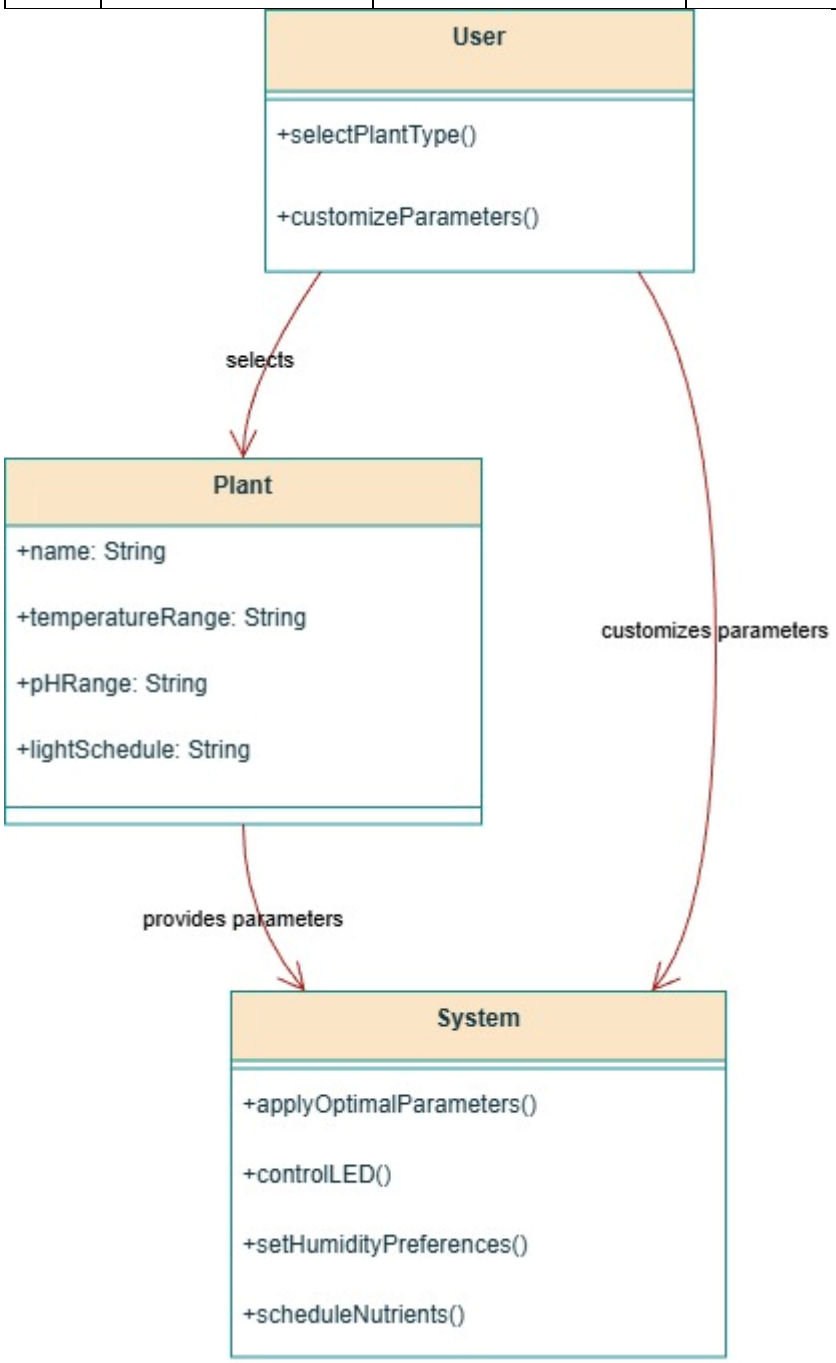
Row	Function / Flow	Actor / Trigger	Steps & Details
1	A. Real-time Dashboard	Primary: Mobile App User / End User Secondary: Cloud Service Trigger: User opens app	1. User opens mobile app or web interface. 2. System displays: • Current sensor readings (color-coded) • System status indicators • Camera feed (future) • Manual override controls • Growth progress photos
2	B. Historical Reports	Trigger: User requests data	1. User requests historical data. 2. System provides: • 24-hour parameter graphs • 7-day trend analysis • Nutrient consumption charts • Alert history log
3	C. Vacation Mode	Trigger: User enables mode	1. User enables vacation mode. 2. System adjusts to: • Increase system check frequency • Send daily status summaries • Enable extended reservoir capacity logic • Prioritize critical alerts


User
+openApp()
+requestHistoricalData()
+enableVacationMode()
opens mobile app or web interface
requests historical data
enables vacation mode
System
+displaySensorReadings()
+displayStatusIndicators()
+displayCameraFeed()
+displayControlButtons()
+displayGrowthPhotos()
+provideHistoricalData()
+increaseCheckFrequency()
+sendDailyStatusSummaries()
+enableExtendedReservoir()
+prioritizeCriticalAlerts()
displays
provides
prioritizes critical alerts
SensorReadings
+colorCoding()
HistoricalData
+graphs24Hour()
+trendAnalysis7Day()
+nutrientConsumptionCharts()
+alertHistory()

UC-6: Configure Plant-Specific Parameters

Row	Flow / Step	Actor / Condition	Action & Details
1	Primary Actor & Trigger	Primary Actor: End User Secondary Actor: Plant Database Trigger: User begins setup or reconfiguration	*(This row sets the context for the use case)*
2	Main Flow: 1. Plant Selection	User selects a plant type from a predefined list.	Options include: • Lettuce: Temp 18-22°C, pH 5.5-6.0, Light 14-16hrs • Spinach: Temp 15-20°C, pH 6.0-6.5, Light 12-14hrs • Kale: Temp 10-25°C, pH 6.0-7.0, Light 14-16hrs • Herbs (e.g., Basil): Temp 20-25°C, pH 5.5-6.5, Light 16-18hrs
3	Main Flow: 2. Apply Parameters	System receives the plant type selection.	System automatically applies the optimal parameters for the chosen plant, including: • Temperature range • pH range • Light schedule (for LED control) • Humidity preferences • Nutrient schedule
4	Main Flow: 3. Customization	User reviews the applied settings.	The user has the option to customize any parameter (e.g., temperature, light hours) to

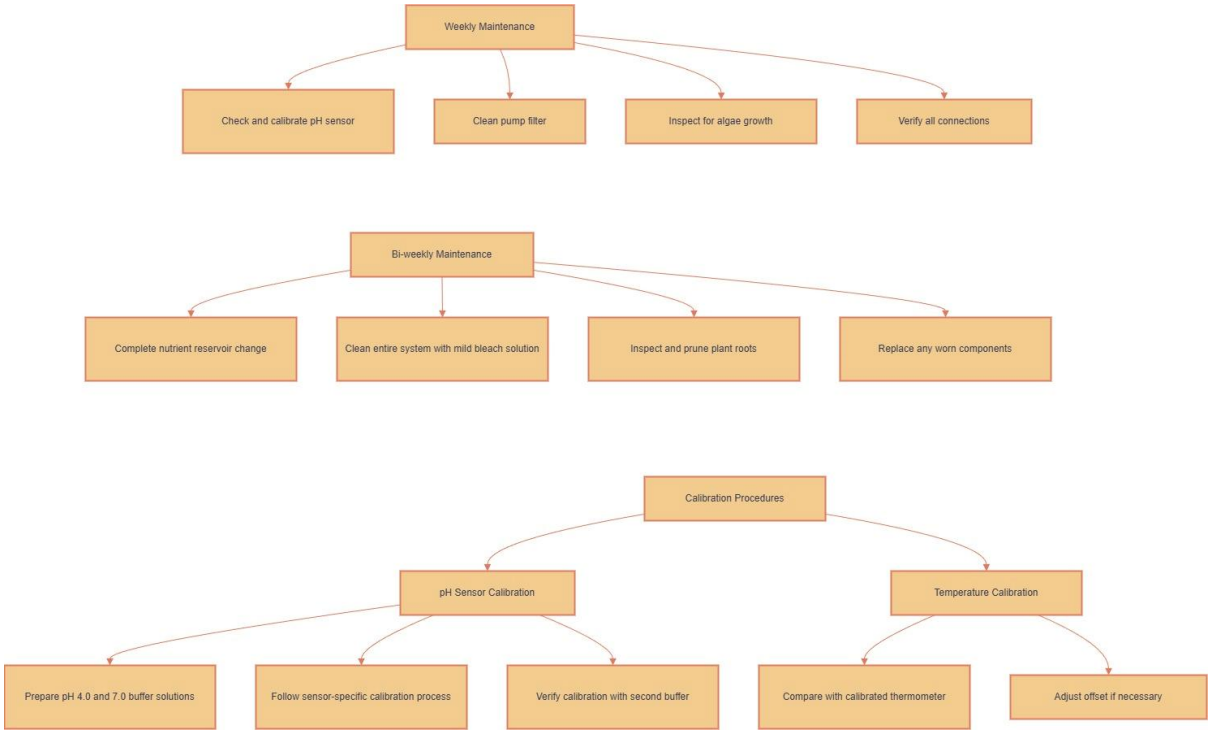
			override the system's default optimal settings.



UC-7: Perform System Maintenance

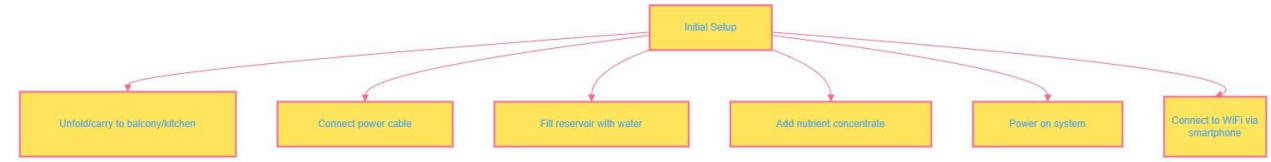
Row	Maintenance Type / Step	Actor / Condition	Actions & Details
1	Primary Actor & Preconditions	Primary Actor: End User Preconditions: System alert OR scheduled maintenance trigger	*(Context for the maintenance activity)*
2	A. Weekly Maintenance	End User performing routine check	1. Check and calibrate pH sensor. 2. Clean pump filter. 3. Inspect system for algae growth. 4. Verify all tubing and electrical connections.
3	B. Bi-weekly Maintenance	End User performing comprehensive refresh	1. Perform complete nutrient reservoir change. 2. Clean entire system with a mild bleach solution. 3. Inspect and prune plant roots if necessary. 4. Replace any worn components (e.g., filters, tubing).
4	C1. pH Sensor Calibration	End User ensuring sensor accuracy	Procedure: 1. Prepare pH 4.0 and 7.0 buffer solutions. 2. Follow the sensor manufacturer's specific calibration process. 3. Verify the calibration by

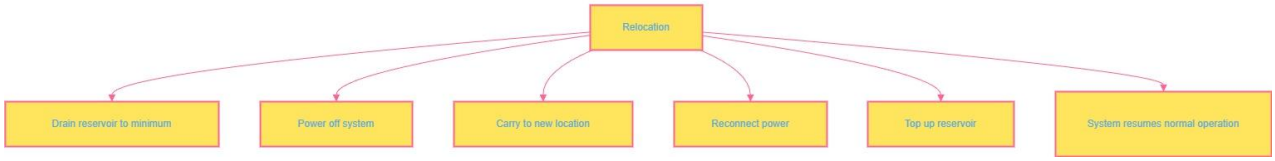
			testing with the second buffer solution.
5	C2. Temperature Calibration	End User verifying temperature readings	Procedure: 1. Compare the system's temperature reading with a known, calibrated thermometer. 2. Adjust the sensor's offset in the system settings if a discrepancy is found.



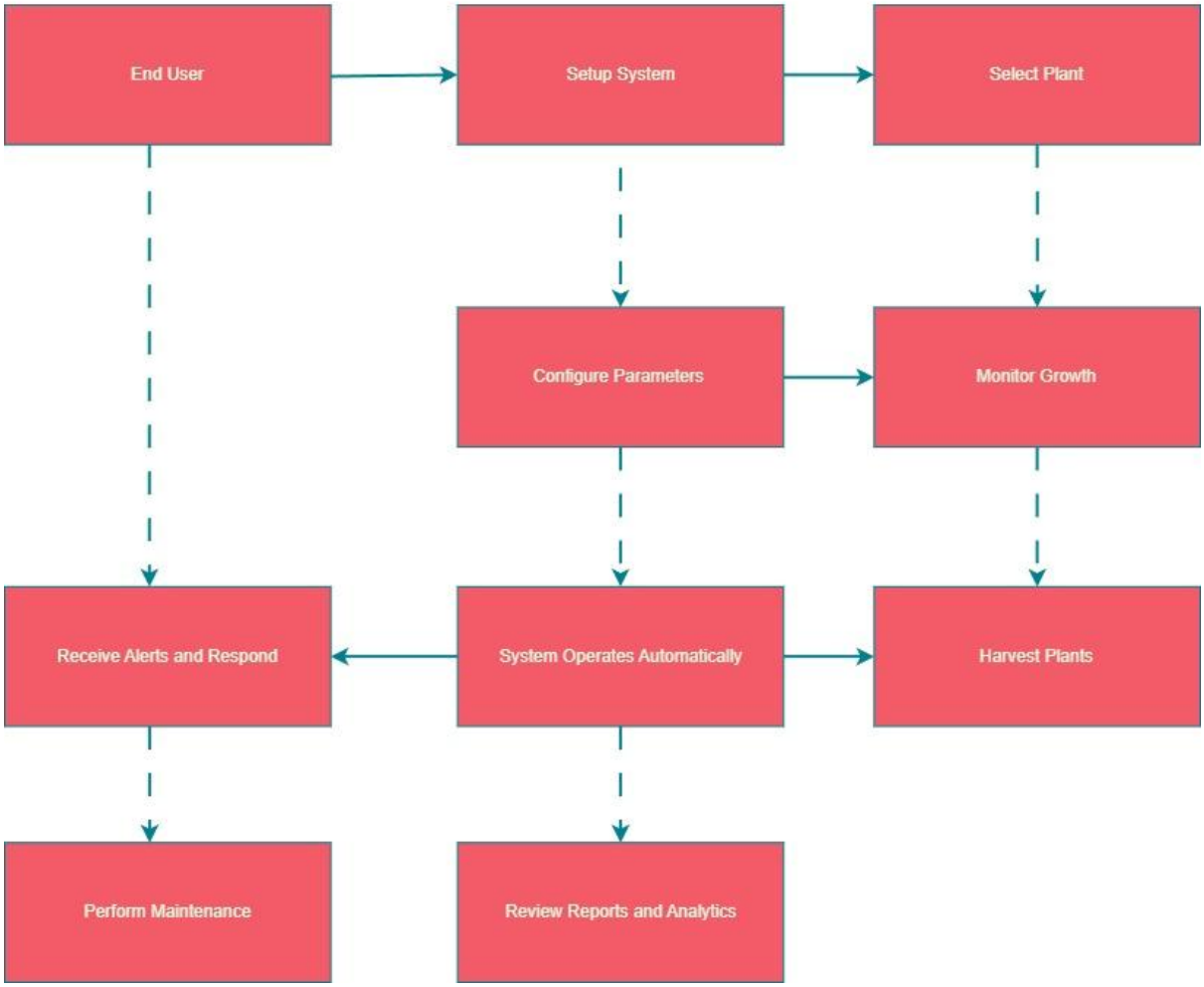
UC-8: Portable Setup & Mobility

Row	Process Phase	Primary Actor / Context	Steps & Details
1	Context	Primary Actor: End User	*(This uses case covers the initial setup and physical relocation of the system.)*
2	Initial Setup	User is preparing the system for first use.	1. Unfold/carry unit to location (e.g., balcony/kitchen). Max dimensions: 60×40×40cm. 2.Connect the power cable. 3. Fill the reservoir with water. 4. Add nutrient concentrate. 5. Power the system on. 6. Connect to Wi-Fi via the smartphone app.
3	Relocation	User needs to move the system to a new location.	1. Drain the reservoir to its minimum level. 2. Power off the system. 3. Carry to new location. Weight when empty: <15kg. 4. Reconnect power. 5. Top up the reservoir with water/nutrients. 6. System resumes normal operation automatically.


Initial Setup
Unfold/carry to balcony/kitchen
Connect power cable
Fill reservoir with water
Add nutrient concentrate
Power on system
Connect to WiFi via smartphone


Relocation
Drain reservoir to minimum
Power off system
Carry to new location
Reconnect power
Top up reservoir
System resumes normal operation

User Interaction Flow



System Component Interactions

SENSORS	ACTUATORS	COMMS	STORANGE	POWER
pH	LED Strips	Wi-Fi	Local Log	12v Input
Temperature	Water Pump	Bluetooth	Cloud Sync	DC-DC Conv
Humidity	Fan	MQTT	OTA Updates	Backup Battery
CO2	Heater	HTTP API		
Water Level				